

Lab 10: Ensemble Learning

Objective:

The purpose of this lab is to study the Ensemble Learning. Students will:

- One dataset is used to learn the objective the Ensemble Learning
- Ensemble Learning methods, including Bagging, Random Forest, AdaBoost, and Gradient Boosting, in comparison with baseline models (Logistic Regression and Decision Tree).
- Display visualizations directly in the notebook for all comparisons.

Software Requirements:

- 1) Python
- 2) scikit-learn pandas
- 3) numpy
- 4) Matplotlib

Dataset:

`sklearn.datasets.breast_cancer_dataset`

Lab Instructions

1. Import necessary libraries.

2. Data Preparation, Visualization, Pre-processing, and Train-Test Split (10 pts)

A. Data Preparation (5 pts)

Load the datasets and perform an initial data exploration. Specifically:

- Display the shape of the dataset
- Print the first few rows to inspect sample entries.
- Visualize feature relationships using `pairplot()`

B. Train-Test Split (5 pts)

Split each dataset into training and testing subsets using a 70%-30% ratio.

Report the shapes of the resulting training and testing sets.

3. Scaling (5 pts)

After performing the train-test split:

- Apply standardization using `StandardScaler`, fitting only on the training set and then transforming both training and testing sets.

4. Set up baseline models (5pts)

Set up baseline models (1) Logistic Regression and (2) Decision Tree to first predict the result, print out the accuracy score.

5. Set up Ensemble models (15pts)

Set up four ensemble models (1) Bagging, (2) Random Forest, (3) AdaBoost, and (4) Gradient Boosting, print out the accuracy score. (10pts)

Written Questions (5pts)

(1) From task 4 and 5, by comparing the baseline models & ensemble models, briefly discuss what have you found?

6. Voting classifier and Stacking classifier (10pts)

Using three models in the voting classifier, and two models for the stacking classifier, models can be selected from the baseline model and the ensemble models, print out the accuracy. (5pts)

Written Questions (5pts)

(1) What is the difference between the voting and stacking classifiers?

7. Random Forest evaluation (50pts)

A. Out-of-Bag (10pts)

For the Random Forest model, print out the OOB score (set the parameter to be "oob_score"). (5pts)

Written Question (5pts)

(1) What is OOB?

B. Bias-Variance (20pts)

For the `n_estimators` in Random Forest classifier, estimate diverse `n` (e.g. `n` from 10 to 200 with step 10), plot out the curve of the train and test accuracy. (10pts)

Written Questions (10pts)

(1) What have you found from the curve plot? (2) What is Bias-Variance trade-off?

C. Confusion matrix (5pts)

Display the confusion matrix using the predicted results obtained from the Random Forest classifier.

D. Feature importance analysis (10pts)

Plot the feature importance of the Random Forest classifier.

E. Learning curve (10pts)

Plot the learning curve of the Random Forest classifier.

Submission Requirement:

- 1) Complete tasks in a jupyter notebook file, name your file as “name_student number.ipynb” and submit it on iSpace.
- 2) Please provide necessary outputs to support your assignment. (e.g. printing necessary information or visualizing through graphs and tables)

Deadline: 23:59, May. 12, 2026, Plagiarism (cheating) = F